P.3 Operations on Functions; Graphing Techniques

Name _____ Period ____ Date _____

AP Calculus P.3 · Student sheet

I can…

- combine functions and find the domain of the result.
- form a composite function, find its domain, and write a function as a composition.
- graph a function by shifting, reflecting, compressing, or stretching a basic graph.

Vocabulary · fill in the missing word

Composite function — $(f \circ g)(x) =$ _____

Shifts — $y = f(x) + k$ moves the graph vertically; $y = f(x - h)$ moves it _____

Reflections — $y = -f(x)$ _____ the graph about the x-axis; $y = f(-x)$ reflects it about the y-axis

Compressions and stretches — $y =$ _____ $= f(ax)$, with $a > 0$, $a \neq 1$

Part 1 · Build a graph in stages

Start with $y = \sqrt{x}$. Add one change at a time.

Equation	What changed?
$y = \sqrt{(x - 2)}$	
$y = \sqrt{(x - 2)} + 1$	
$y = -\sqrt{(x - 2)} + 1$	
$y = 2\sqrt{(x - 2)} + 1$	

Part 2 · Notes and examples (fill in each note, then work the example)

Sum, difference, product, quotient: the domain is the numbers in the domains of both f and g; for f/g, also $g(x) \neq$ _____.

Example 1

$f(x) = \sqrt{x + 1}$ and $g(x) = \frac{2}{x}$. Find $(f + g)(x)$ and $(\frac{f}{g})(x)$ and the domain of each.

Domain of $f \circ g$: the numbers x in the domain of g for which g(x) is in the domain of _____.

Example 2

$f(x) = x^2 + 1$ and $g(x) = \sqrt{x - 1}$. Find $f \circ g$ and $g \circ f$ and the domain of each.

Example 3

Find functions f and g so that $f \circ g = F$.

a. $F(x) = (2x + 3)^4$

b. $F(x) = \sqrt{x^2 + 1}$

Shifts: $y = f(x - h) + k$ shifts the graph of f right h units and _____ k units (for $h, k > 0$).

Example 4

Graph $f(x) = (x - 1)^3 + 2$ starting from $y = x^3$. Where does the point (0, 0) go?

Stretches: $y = af(x)$ stretches the graph vertically when $a > 1$ and _____ it when $0 < a < 1$.

Example 5

Describe how to get $g(x) = -4\sqrt{x - 1}$ from $y = \sqrt{x}$, and give two points on g.

Part 3 · Partner practice

Partner 1

$f(x) = \dfrac{x}{x - 1}$ and $g(x) = \dfrac{2}{x}$. Find $(f \circ g)(x)$ and its domain.

Partner 2

Describe how to get $G(x) = 2|x - 1|$ from $y = |x|$.